

Ritweek Raj

Roll No.: U24AI067

Bachelor of Technology

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🐙 GitHub Profile

🌐 LinkedIn Profile

🔗 LeetCode

🌐 Portfolio

EDUCATION

•Bachelor of Technology in Artificial Intelligence

2024 – 2028

Sardar Vallabhbhai National Institute of Technology, Surat

CGPA: 9.22

•Class XII(CBSE Board)

2022 – 2023

St. Karen's Secondary School, Patna

Percentage: 96.4%

•Class X(CBSE Board)

2020 – 2021

St. Karen's Secondary School, Patna

Percentage: 98%

PROJECTS

•SmartShop – E-commerce Smart Search & Relevance Optimizer

Live Project

AI-powered semantic search engine for e-commerce

- Built a full-stack e-commerce search platform using React.js, Node.js, Express.js, FastAPI, MongoDB, and Elasticsearch.
- Developed an NLP-powered search engine using Sentence-BERT, spaCy, and RapidFuzz supporting natural language queries, typo correction, synonym expansion, and semantic search.
- Implemented hybrid search by combining BM25 keyword ranking, KNN vector search, and business score boosting for accurate product retrieval.

•AI Image Detector

Live Project

Vision Transformer based AI-generated image detection

- Fine-tuned a Vision Transformer (ViT) using PyTorch and Hugging Face Transformers to classify AI-generated and real images.
- Built a FastAPI inference API with efficient image preprocessing and low-latency prediction.
- Developed a responsive React.js frontend integrated with the ML backend for real-time image classification.

TECHNICAL SKILLS

Languages: Python, JavaScript (ES6+), TypeScript, HTML5, CSS3

Frontend: React.js, Next.js, Tailwind CSS

Backend: Node.js, Express.js, FastAPI, REST APIs

Databases: MongoDB, Elasticsearch

Machine Learning: PyTorch, Hugging Face Transformers, Vision Transformer (ViT), Sentence-BERT (SBERT), NLP, spaCy, RapidFuzz

Tools: Git, GitHub, Docker, Kaggle, Jupyter Notebook, Vercel, Render

ACHIEVEMENTS

•Winner – Web Wonders 1st place in NEXUS Web Development Competition

Sep 2025

•Top 3 Finalist – Genesis Innovation and Problem-Solving Competition

Jan 2026

•2nd Runner-Up – Mr. Gujarat 2026

Mar 2026

POSITIONS OF RESPONSIBILITY

•Event Anchor Anchored PowerMind, Teachers' Day Celebration, and First-Year Orientation (Trivia Session)

•DSA Mentor Mentored junior students in Data Structures and Algorithms

•Event Management – NEXUS Organized and managed Riddle Fuse event operations